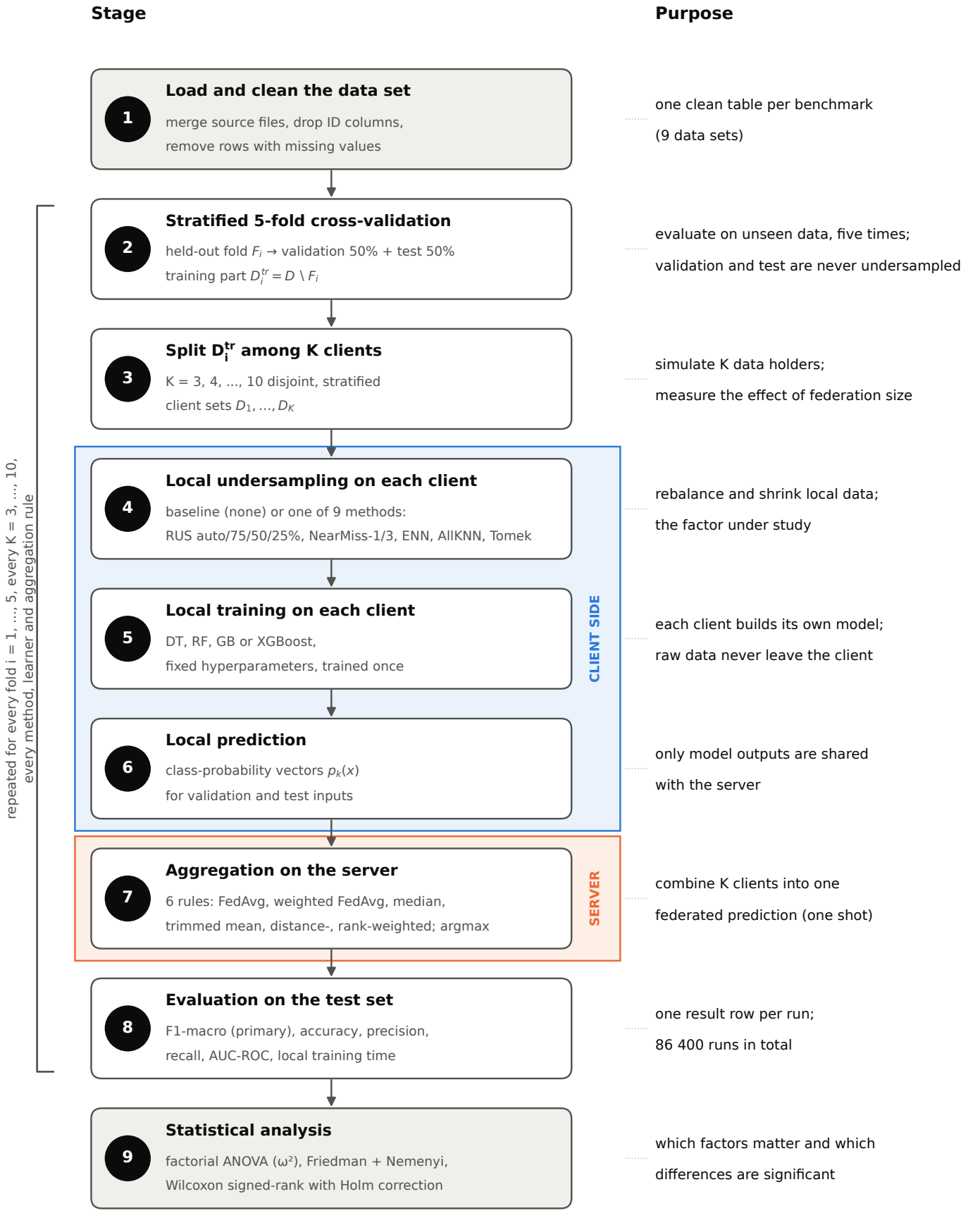
Stage
Purpose
1
Load and clean the data set
merge source files, drop ID columns,
remove rows with missing values
one clean table per benchmark
(9 data sets)
2
Stratified 5-fold cross-validation
held-out fold F_i → validation 50% + test 50%
training part $D_i^{tr} = D \setminus F_i$
evaluate on unseen data, five times;
validation and test are never undersampled
3
Split D_i^{tr} among K clients
K = 3, 4, ..., 10 disjoint, stratified
client sets $D_1, \ldots, D_K$
simulate K data holders;
measure the effect of federation size
4
Local undersampling on each client
baseline (none) or one of 9 methods:
RUS auto/75/50/25%, NearMiss-1/3, ENN, AllKNN, Tomek
rebalance and shrink local data;
the factor under study
5
Local training on each client
DT, RF, GB or XGBoost,
fixed hyperparameters, trained once
each client builds its own model;
raw data never leave the client
6
Local prediction
class-probability vectors $p_k(x)$
for validation and test inputs
only model outputs are shared
with the server
CLIENT SIDE
7
Aggregation on the server
6 rules: FedAvg, weighted FedAvg, median,
trimmed mean, distance-, rank-weighted; argmax
combine K clients into one
federated prediction (one shot)
SERVER
8
Evaluation on the test set
F1-macro (primary), accuracy, precision,
recall, AUC-ROC, local training time
one result row per run;
86 400 runs in total
9
Statistical analysis
factorial ANOVA (ω²), Friedman + Nemenyi,
Wilcoxon signed-rank with Holm correction
which factors matter and which
differences are significant
repeated for every fold i = 1, ..., 5, every K = 3, ..., 10,
every method, learner and aggregation rule